

Final Report

The goal of this project was to redesign the Montgomery Group research website into a cleaner, more modern, and easier-to-maintain site. The main sections of the website include Home, People, Publications, Projects, Resources, and Highlights. By the end of the semester, the main structure of the website was complete, the visual design was more consistent, and most of the important content had been migrated and updated.

Since the midterm, the project made strong progress. The People section became much more complete as I added more group member information, reorganized the page into clearer sections, and improved the individual profile pages. The Publications section was also expanded with more entries, images, abstracts, and external links such as DOI, PubMed, PubMed Central, and Semantic Scholar. I also changed the publication data into a structured JSON format, which made the site easier to update and more consistent with the People and Projects sections. The Projects section was refined by improving the project detail pages, correcting image issues, and making the overall layout more polished. The Resources and Highlights sections also improved and were reorganized so they can use JSON data files for future updates.

Another important part of the project was documentation. Because the final report alone is not enough for complete documentation, I also prepared an updated README, a simple update guide for Dr. Montgomery, and developer notes for future maintainers. These files explain how the site is organized, how normal updates should be made, and how future developers can continue extending the site.

At this point, the core project is mostly complete. The main structure, design, and most of the key content are finished. What remains is mainly future content expansion rather than major development. For example, more highlights can be added later, more resources can be included over time, and publication entries can still be expanded with more images or

links if needed. In other words, the main redesign and implementation work is finished, and the site can now continue to grow through maintenance.

One challenge that continued after the midterm was collecting complete information for the People page. Some of the content depended on receiving photos and biographies from group members, so this could not be solved only through coding. I addressed this by updating the page gradually as new information became available.

Another challenge was deciding how to make the site easier for non-technical users to maintain. A more advanced editing system similar to Owlstown was considered, but creating a full content management system would have been far beyond the scope of this project. Instead, I addressed this by moving repeated content into structured data files and writing clear documentation. This was a simpler and more realistic solution.

A third challenge was keeping the design consistent as more content was added. New layout issues appeared over time, especially with image display, spacing, and page balance. I addressed these issues through repeated testing, CSS revisions, and supervisor feedback. The migration process also continued to take time because the old website content could not be moved directly and had to be reorganized manually.

The overall goals of the project did not fundamentally change. The website was still intended to become a cleaner, more organized, and easier-to-maintain research group website. However, one important shift happened during the second half of the semester. Earlier in the semester, the main focus was on redesign and content migration. Later, future maintenance became a more important part of the project. Because of this, I placed more emphasis on structured data files, simpler update workflows, and written documentation. This did not replace the original goals, but it refined them.

This project was completed as an individual project. Therefore, all work was completed by me.

Member contribution: Longyu Tang - 100%

Major tasks completed:

- weekly meetings with the supervisor and review of feedback
- planning the website structure
- reviewing the original website
- setting up the development environment
- building and revising the website framework
- migrating and reorganizing content
- improving the People, Publications, and Projects sections
- updating detail pages and publication data
- refining the Resources and Highlights pages
- preparing project documentation
- final testing, cleanup, and repository organization

Estimated total time spent over the full semester: approximately 128 hours

Estimated average time spent per week: approximately 8–9 hours

If this project is handed off to another student or future maintainer, the next person should begin by reading the README, the update guide for Dr. Montgomery, and the developer notes. These documents explain how the site is organized and how normal updates should be made.

Most future work should focus on maintenance rather than rebuilding. The People, Projects, and Publications sections are already organized around structured data files, so future updates should continue using that system. In most cases, the next maintainer should update JSON files and images rather than rewriting page layout.

The next likely areas of expansion are Highlights and Resources. These pages now have a better structure, but they still have less content than the main sections. If they continue

to grow, they should follow the same design pattern used elsewhere in the site: one page template, one data file, and simple JavaScript to load repeated content. If the research group later decides that it wants a full editing system similar to Owlstown, that would be a separate and much larger project. For now, the site should continue to be maintained through structured data files and documentation.

GitHub Repository:

<https://github.com/LongyuT/research-website>

The Work Log

Week 1:

Met with supervisor and reviewed the current Owlstown website, discussed project goals and identified the main pages of the new website – 1 hour

Reviewed the structure and content of the existing website, analyzed its strengths and weaknesses, and explored alternative approaches to structuring the site. – 1 hour

Total – about 2 hours

Week 2:

Researched possible hosting platforms and deployment options – 1.5 hours

Compared different methods for updating and maintaining the website – 1 hour

Discussed domain name and update workflow with supervisor – 0.5 hour

Total – about 3 hours

Week 3:

Set up development environment and project structure – 2 hours

Configured deployment workflow and connected the repository – 2 hours

Created initial website framework and navigation layout – 6 hours

Built the first demo homepage – 3 hours

Total – about 13 hours

Week 4:

Started migrating home and publication content from the original website – 7 hours

Rebuilt homepage content and reorganized text sections – 4 hours

Created initial structure for the People page – 2 hours

Adjusted navigation layout and page links – 2 hours

Total – about 15 hours

Week 5:

Continued migrating publications from the old website – 6 hours

Reformatted publication entries to fit the new layout – 4 hours

Created the Resources page and added initial structure – 3 hours

Minor layout adjustments for consistency – 2 hours

Total – about 15 hours

Week 6:

Tested different CSS layout and color schemes – 6 hours

Improved visual consistency across pages – 3 hours

Adjusted publication layout based on supervisor feedback – 2 hours

Refined spacing and alignment across the website – 3 hours

Total – about 14 hours

Week 7:

Updated banner image and homepage design – 2 hours

Improved layout consistency across all pages – 2 hours

Continued cleaning and adjusting migrated content – 5 hours

Fixed minor formatting and styling issues – 2 hours

Total – about 11 hours

Week 8:

Updated the People page with additional group member information and reorganized sections for current members and former members – 2 hours

Added photo handling logic and improved person profile pages – 1 hours

Total – about 3 hours

Week 9:

Added more publication entries, images, and detail-page content – 4 hours

Improved publication layout and external links for papers – 4 hours

Tested detail pages and corrected formatting issues – 2 hours

Total – about 10 hours

Week 10:

Expanded Projects page content and improved project image layout – 4 hours

Fixed image sizing inconsistencies and refined CSS for project display – 4 hours

Updated project detail page structure – 2 hours

Total – about 10 hours

Week 11:

Added more people information and undergraduate researcher entries – 2 hours

Updated Resources page and Highlights page – 1 hours

Continued site-wide testing and cleanup – 1 hours

Total – about 4 hours

Week 12:

Converted publication content into a cleaner JSON-based structure – 4 hours

Updated publication page scripts to use the new data source – 4 hours

Fixed homepage recent publications section to use the same data – 2 hours

Total – about 10 hours

Week 13:

Prepared README and project documentation – 4 hours

Wrote update instructions for the supervisor – 3 hours

Wrote developer notes for future maintainers – 3 hours

Total – about 10 hours

Week 14:

Expanded support for Resources and Highlights as data-driven pages – 2 hours

Tested final site behavior and fixed page-level issues – 3 hours

Reviewed repository structure and update workflow – 3 hours

Total – about 8 hours

Estimated total: about 128 hours